

FEDERICO COLOMBO

Computer Graphics Engineer | Technical Artist | XR Developer

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PROFESSIONAL PROFILE

Computer Graphics Engineer and Technical Artist specialized in real-time rendering, spatial computing (XR), and procedural tool development. Graduated with honors in Cinema and Media Engineering. Deep expertise in writing custom C++ rendering engines, optimizing GPU architectures (HLSL/GLSL), and leading multidisciplinary teams. Actively seeking to leverage hardcore engineering and project management skills to build scalable software architectures and high-fidelity interactive productions.

EDUCATION

M.Sc. Cinema and Media Engineering - Politecnico di Torino 2024 - 2026

- **Expected Grade:** 110/110 Cum Laude (Graduating Dec 2026)

B.Sc. Cinema and Media Engineering - Politecnico di Torino 2021 - 2024

- **Final Grade:** 100/110

WORK EXPERIENCE

Lemons In The Room & Politecnico di Torino | Turin / Florence, Italy 03/2026 - Present

XR Master Thesis - R&D Graphics Programmer (PulseEngine)

- Engineered **PulseEngine**, a real-time bio-adaptive telemetry and volumetric CSG framework in Unity 6 for standalone Medical XR on Meta Quest 3.
- Architected a **Virtual DAW Mixer Console** (`DAWMixerWindow`) inside Unity featuring channel strips, live OLED oscilloscopes, VU meters, and automated packet sniffing to route multi-sensor biometric streams (BLE, OSC, WebSockets, Serial).
- Developed a GPU-evaluated volumetric CSG pipeline with dynamic blend softness; eliminated the 180° normal inversion symmetry seam bug in triplanar mapping using the **Morten Mikkelsen Surface Gradient** formulation ($\nabla_{\text{surf}}(h) = \nabla h - (N \cdot \nabla h)N$), guaranteeing continuous C^1 micro-relief.
- Engineered a Compute Shader GPU voxelizer baking 128^3 distance grids in under 0.8ms, powering **Physics-Only Mode**: decoupled GPU fill-rate for VFX Graph particle adhesion to achieve a locked **90 FPS on standalone Quest 3** (2.91ms GPU time, 0B GC alloc) with a 61.3% GPU load reduction.
- Integrated affective biofeedback mapping autonomic stress along the neurological **Bouba/Kiki spectrum**; modularized the architecture to official Unity Package (UPM) standards for

commercial release on the Unity Asset Store.

Level Up Lab | Turin, Italy

01/2025 - Present

Technical Artist & Gameplay Programmer

- Programmed core gameplay mechanics in C# (Unity) and designed the technical architecture for the "Line Operator" project.
- Acted as Project Manager and Technical Producer, coordinating multidisciplinary teams of designers and artists to oversee artistic vision and technical viability.
- Developed custom HLSL Shaders and post-processing pipelines to establish the game's unique visual identity.

Politecnico di Torino | Turin, Italy

10/2024 - 01/2025

Teaching Assistant (GUI)

- Tutored students in UI/UX principles and frontend programming during weekly laboratory sessions.
- Wrote technical documentation and created high-quality instructional content to improve comprehension.

RKH SRL | Turin, Italy

09/2023 - 06/2024

Video Artist

- Operated professional cameras (RED, Blackmagic 4K/6K) and assisted DOP with complex lighting setups.
- Handled editing and post-production workflows using DaVinci Resolve for corporate content.

SKILLS

- **Languages:** C++, C#, Python, HLSL/GLSL
- **Engines:** Unity, Unreal Engine 5, Custom OpenGL Frameworks
- **Software:** Blender, DaVinci Resolve, Jira, Git
- **Management:** Project Management, Technical Direction, Agile Product Strategy

PERSONAL PROJECTS

- **Smart Cable Generator (Blender Tool):** Engineered a Python add-on for Blender to procedurally generate parametric hanging cables across multi-object modular assets with scale-independent catenary physics and automated mesh conversion for game engines.
- **Graphics Lab (Custom C++ Engine):** Engineered a custom OpenGL 3D rendering framework from scratch. Implemented Deferred Shading, Physically Based Rendering (PBR), and Silhouette Parallax Occlusion Mapping (SPOM) to study low-level GPU pipelines.
- **UE5 Tech Art Pipeline:** Developed procedural environments using PCG. Scripted a homing projectile system in Niagara and integrated MetaHumans with custom groom collision physics.

- **Character Rigging & Tooling:** Built complete custom FK/IK character rigs (Blender), painting skin weights and setting up constraint networks. Developed Python automation scripts to streamline 3D asset export pipelines into game engines.
- **Awaken Asylum & Nihil:** Developed an immersive VR educational experience (Unity) and a 2D Metroidvania prototype emphasizing complex transformation mechanics. Managed end-to-end technical production.